

# Fair Ranking in Retrieval-Augmented Generation: Is the Benefit Fairness, or Diversity in Disguise?

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## Abstract

Kim and Diaz (Kim and Diaz, 2025) report that Retrieval-Augmented Generation (RAG) systems built on item-exposure-fair rankings can preserve, and in many cases even improve, downstream generation quality relative to relevance-only rankings. We investigate a specific confound in that claim: the stochastic fairness mechanism most commonly used for this purpose, Plackett-Luce (PL) reranking, does not only equalize expected exposure across repeated rankings — as a byproduct of its randomization, it also tends to change the within-list textual diversity of what a RAG generator actually sees. If diversity itself helps generation, an apparent “fairness helps” result could really be diversity doing the work. We test this directly on the LaMP benchmark (all 7 tasks, both generators Flan-T5-Small and Flan-T5-Base, both BM25 and Contriever retrieval, full query coverage, 336 experiment configurations, 163,392 query-level observations) by using Maximal Marginal Relevance (MMR) — a reranker with an explicit diversity dial and no fairness objective — to separate the two constructs. We find no evidence that PL reranking improves utility over the deterministic baseline (10 of 16 Holm-corrected tests are significant, and 8 of those are losses); that within PL’s own fairness dial, greater exposure equality is robustly associated with *lower* utility (coefficient +0.0735, cluster-robust  $p = 6.3 \times 10^{-54}$ , stable under leave-one-task-out and three levels of fixed effects); that MMR-induced diversity shows no general utility benefit of its own (a modest, heterogeneous negative pooled association,  $p = 0.0115$ ); and that PL retains an average utility disadvantage relative to MMR after adjusting for measured diversity (average marginal effect  $\approx -0.019$  to  $-0.020$ , robust across every specification and every leave-one-task-out fold we tested). A directly tested candidate mechanism — that PL pays a relevance cost via reduced exposure-relevance alignment — does not explain this residual. We conclude that, in the settings we test, measured within-list diversity (ILD-Jaccard) alone does not account for the difference between PL and MMR utility, while explicitly declining to interpret this residual as the causal cost of fairness itself, since the two rerankers differ along dimensions besides diversity.

## 1 Introduction

Retrieval-Augmented Generation (RAG) systems answer a query by first retrieving a small set of supporting documents or profile items and then conditioning a language model’s generation on that retrieved set (Lewis et al., 2020). Because only a handful of items are shown to the generator, *which* items are retrieved and in *what order* they are presented has a direct effect on the generator’s output. This same property has made ranking a natural place to intervene for *item-side fairness*: if being retrieved and shown carries value (exposure) to the entity that produced an item, then systematically favoring some relevant items over other, comparably relevant items is a fairness concern independent of whether it also affects end-task quality. A substantial ranking-fairness literature formalizes this as exposure fairness — equalizing the *expected exposure* an item receives across repeated, possibly randomized rankings, rather than requiring a single fixed ranking to look fair on its own (Singh and Joachims, 2018; Biega et al., 2018; Diaz et al., 2020).

Kim and Diaz (Kim and Diaz, 2025) apply this idea to RAG directly, reranking retrieved candidate sets with fairness-aware methods before generation and measuring both item-side exposure fairness and downstream generation quality. Their central, and somewhat counter-intuitive, finding is that RAG systems built on fair rankings can preserve, and in many cases even outperform, RAG systems built on purely relevance-ranked retrieval — suggesting that enforcing fairness need not cost generation quality, and might sometimes help it.

This project investigates a specific reason that result could be misleading. The fairness mechanism most commonly used in this line of work, Plackett–Luce (PL) stochastic reranking, achieves exposure equality by randomizing which comparably relevant items are shown across repeated draws (Plackett, 1975; Luce, 1959). That randomization is not targeted at diversity in any direct sense, but as a side effect it tends to change how varied the items shown *in any single sampled list* are from one another. If a RAG generator produces better output when it sees a more varied context — independent of whether that context is fairly exposed — then an observed “fair reranking helped” result could really be within-list diversity doing the work, with fairness merely along for the ride. Before attributing a utility change to fairness, it is therefore necessary to check whether the diversity that fairness happens to induce is the actual mechanism.

This is the question this paper answers:

*When a fair ranking method appears to help, or hurt, a RAG system’s output quality, is that effect attributable to fairness itself, or to the within-list diversity that the fairness mechanism happens to produce?*

Answering this requires separating two constructs that are easy to conflate in practice but are conceptually distinct: *fairness*, which we operationalize as expected exposure equality across repeated rankings (Expected Exposure Disparity, EE-D, lower is fairer), and *diversity*, which we operationalize as within-list textual variety in a single ranking (ILD-Jaccard). PL moves both at once. What lets us pull them apart is Maximal Marginal Relevance (MMR) (Carbonell and Goldstein, 1998): a deterministic reranker with an explicit relevance-diversity dial and *no* fairness objective whatsoever. Comparing PL to MMR after adjusting for measured diversity level can reveal whether PL’s outcomes track diversity alone or whether a residual difference remains — though, as we discuss throughout, that residual is not automatically identifiable as “the causal effect of fairness,” since PL and MMR differ along several dimensions besides diversity (stochastic versus deterministic selection, how each preserves the underlying relevance ordering, and so on).

**Contributions.** We (1) reproduce and re-examine Kim and Diaz’s central claim on a full-coverage subset of their experimental design (all 7 LaMP tasks, two generators, two retrievers, complete query coverage); (2) test, rather than assume, whether PL’s fairness mechanism actually induces the diversity increase the confound requires, at both the aggregate and per-query level; (3) use MMR as an explicit diversity-only instrument to separate the fairness and diversity constructs empirically, reporting four main findings with cluster-robust inference, leave-one-task-out robustness, and three progressively stricter levels of fixed-effects identification; and (4) directly test and reject our own leading hypothesis for the residual PL–MMR utility gap (a relevance cost from PL’s stochastic sampling), leaving the mechanism explicitly open rather than overclaiming an explanation.

## 2 Background and Related Work

**Retrieval-augmented generation.** RAG conditions a generation model on a small set of retrieved documents or profile items rather than relying solely on parametric knowledge (Lewis et al.,

2020). This makes generation quality directly sensitive to the retrieval and reranking pipeline: which items are retrieved, and the order and composition of the final shortlist handed to the generator, can change what the model is able to produce. Reranking the initial candidate set — the stage this paper studies — is therefore a natural place to intervene on both fairness and diversity objectives without retraining the generator itself.

**Exposure fairness in ranking.** A ranking necessarily privileges higher-ranked items with more user attention; item-side fairness concerns whether comparably relevant items receive comparable exposure over repeated exposure to the ranking, not whether any single ranking looks balanced in isolation (Singh and Joachims, 2018; Biega et al., 2018). Diaz et al. (2020) formalize this with the *expected exposure* framework: exposure and relevance are both expressed as distributions over items, aggregated across a distribution of (possibly randomized) rankings, and a ranking policy’s disparity is the divergence between the exposure it delivers and the exposure a relevance-proportional target would deliver. This paper’s fairness metric, EE-D (Expected Exposure Disparity), and its companion EE-R (Expected Exposure Relevance, used only in the mechanism test of Section 5.4), are computed with the reference implementation released alongside that framework, applied unmodified to this project’s rankings.

**Kim and Diaz’s Fair RAG.** Kim and Diaz (2025) evaluate nine RAG configurations across the fairness metrics above and downstream task utility, using the LaMP personalization benchmark (Salemi et al., 2024) augmented with their own item-level utility labels. Their central finding — that fair rerankings can preserve or even improve generation quality relative to purely relevance-ranked retrieval — motivates the present paper’s question: whether that apparent quality preservation is attributable to fairness itself, or to a byproduct of the fairness mechanism used to achieve it.

**Plackett–Luce stochastic ranking.** The Plackett–Luce (PL) model (Plackett, 1975; Luce, 1959) defines a distribution over permutations of a candidate set by sampling without replacement: at each step, an unranked item  $i$  with (temperature-scaled) score  $s_i$  is selected next with probability  $\exp(s_i) / \sum_{j \in R} \exp(s_j)$ , where  $R$  is the set of not-yet-selected items. Repeated draws from this distribution, rather than a single deterministic ranking, are what makes expected-exposure equality achievable: items with comparable scores receive comparable exposure *in expectation* even though any single sampled ranking still favors higher-scored items. PL is the fairness mechanism this paper studies; its temperature parameter, denoted  $\alpha$  here, controls how close the sampling distribution is to the deterministic (highest-score-first) ranking, with larger  $\alpha$  producing less randomization and therefore less fairness intervention.

**Maximal Marginal Relevance.** MMR (Carbonell and Goldstein, 1998) constructs a ranking greedily: having selected items  $D_1, \dots, D_{k-1}$ , the next item is

$$D_k = \arg \max_{D_i \notin \{D_1, \dots, D_{k-1}\}} \left[ \lambda \cdot \text{Sim}_1(D_i, Q) - (1 - \lambda) \cdot \max_{D_j \in \{D_1, \dots, D_{k-1}\}} \text{Sim}_2(D_i, D_j) \right],$$

where  $\text{Sim}_1$  is relevance to the query and  $\text{Sim}_2$  is similarity to already-selected items. The parameter  $\lambda \in [0, 1]$  trades relevance for diversity:  $\lambda = 1$  recovers the purely relevance-ranked list, and decreasing  $\lambda$  pushes the ranking toward more mutually dissimilar items. Because MMR is entirely deterministic and has no exposure-equality objective, it gives this paper a diversity manipulation with no fairness confound, which is what makes separating fairness from diversity possible at all.

## 3 Methodology

### 3.1 Datasets

We use the LaMP personalization benchmark (Salemi et al., 2024), all seven tasks, under the “user” split, drawing the full available query set per task from the generator-dependent, per-generator-filtered utility-label test collections that Kim and Diaz’s released data pipeline builds from the original LaMP release (Kim and Diaz, 2025); we do not draw any additional subsample on top of that filtering.

### 3.2 Generators

We evaluate two instruction-tuned Flan-T5 generators (Chung et al., 2024; Raffel et al., 2020): Flan-T5-Small (77M parameters) and Flan-T5-Base (250M parameters). Larger Flan-T5 variants exist in the source paper’s evaluation but were not computationally feasible on the hardware available for this project (Section 8).

### 3.3 Retrievers

We use two retrievers with different underlying representations: a sparse lexical retriever, BM25 (Robertson and Zaragoza, 2009), and an unsupervised dense retriever trained with contrastive learning, Contriever (Izacard et al., 2022).

### 3.4 Rerankers

For each (generator, retriever) cell, we evaluate 12 reranking settings over a top- $k = 5$  candidate list:

- **Deterministic baseline:** the top-5 items by raw retrieval score, with no fairness or diversity intervention. This is the relevance-only reference condition throughout the paper.
- **MMR**, at seven diversity levels  $\lambda \in \{0.15, 0.30, 0.45, 0.55, 0.70, 0.85, 1.00\}$ . As a construction check,  $\lambda = 1$  should exactly reproduce the deterministic baseline’s outcome (Section 4).
- **PL**, at four fairness strengths  $\alpha \in \{1, 2, 4, 8\}$  (lower  $\alpha$  is a stronger fairness intervention), with  $N = 10$  independently sampled rankings per  $\alpha$  per query. Kim and Diaz estimate their stochastic-ranking quantities from  $N = 100$  samples; we use  $N = 10$  as an explicit, disclosed computational compromise (Section 8).

### 3.5 Metrics

**Fairness (EE-D, lower is fairer).** We use Kim and Diaz’s own source-normalized EE-D, computed by their vendored implementation of the expected-exposure framework (Diaz et al., 2020) with `normalize=True` (i.e., normalized against theoretical per-query exposure bounds), applied unmodified to our rankings. We do *not* apply any further normalization to EE-D; wherever  $\text{EE-D}_{\text{norm}}$  appears in this paper it refers to this same source-normalized quantity, not raw EE-D.

**Utility (EU).** Task-specific: accuracy for LaMP-1/2, mean absolute error (MAE, lower is better) for LaMP-3, and ROUGE-L for LaMP-4–7.

**Diversity (ILD-Jaccard).** One minus the mean pairwise Jaccard similarity of the retrieved profile text within a single ranked list; higher is more diverse.

**Exposure-relevance alignment (EE-R).** Also computed by the vendored expected-exposure implementation with `normalize=True`; used only as a candidate mechanism variable in Section 5.4.

### 3.6 Normalization

Because LaMP’s seven tasks use three different utility metrics on different scales, raw EU values are not comparable across tasks. We define our own within-query normalized utility,  $EU_{\text{norm}}$ , distinct from Kim and Diaz’s separate empirical-upper-bound normalization: for query  $q$  and setting  $s$ ,

$$EU_{\text{norm}}(q, s) = \frac{EU'(q, s)}{\max_{s' \in S(q)} EU'(q, s')}, \quad \text{where} \quad EU'(q, s) = \begin{cases} 4.0 - EU(q, s), & \text{LaMP-3} \\ EU(q, s), & \text{otherwise} \end{cases}$$

(LaMP-3 uses MAE, where lower is better, so it is flipped before normalizing; every other task’s raw EU is used as-is) where  $S(q)$  is the set of 12 settings evaluated for query  $q$ ; this is true max-normalization (division by the largest raw value a query achieved), not min-max scaling.  $ILD_{\text{norm}}$  and  $EE-R_{\text{norm}}$  are defined identically, dividing by each query’s own maximum raw ILD or EE-R value across its 12 settings.  $EE-D_{\text{norm}}$  is the one exception: it is never further normalized, since it is already the source-normalized quantity described above.

The normalization denominator is exactly zero, and  $EU_{\text{norm}}$  undefined, precisely when a query scores raw EU=0 under *every one* of its 12 settings — verified directly for all 237 affected queries, with no exceptions. This excludes 5,688 of 163,392 query-level rows (3.5%), concentrated in LaMP-1, 2, 5, and 6 (zero excluded rows in LaMP-3 and LaMP-7); these are queries on which no method received any credit, not queries where methods tied at an informative nonzero value. As a sensitivity check, we re-ran the three primary regressions (Findings 2–4) with these rows filled at  $EU_{\text{norm}}=0$  rather than excluded: coefficients moved modestly and no conclusion changed sign or significance status (Finding 2:  $+0.0735 \rightarrow +0.0697$ ,  $p < 10^{-50}$ ; Finding 3:  $-0.0675 \rightarrow -0.0838$ ,  $p = 0.012 \rightarrow 0.0017$ ; Finding 4:  $-0.0193 \rightarrow -0.0188$ ,  $p < 10^{-33}$ ). We report the exclusion version throughout as the more conservative choice.

### 3.7 Statistical Methodology

**Paired tests.** Wilcoxon signed-rank tests on per-query  $EU_{\text{norm}}$ , paired against the deterministic baseline, Holm-corrected within each (generator, retriever) cell’s test family.

**Cluster-robust inference.** Because each query is observed under multiple settings (every  $\alpha$ , every  $\lambda$ , both methods), we report CR1 sandwich-estimator standard errors clustered by query wherever a pooled regression is the primary estimate, alongside the naive (non-clustered)  $p$ -value for comparison where informative.

**Leave-one-task-out (LOTO).** For every headline pooled coefficient, we re-fit the identical specification after dropping each of the seven LaMP tasks in turn, checking whether the coefficient’s sign or significance status changes.

**Fixed effects.** To separate within-query treatment variation (e.g.,  $\alpha$  changing for the same query) from between-query composition (some queries are systematically easier, fairer-scoring, or more diverse regardless of setting), we estimate two nested fixed-effects specifications via the within (demeaning) estimator with cluster-robust standard errors: a *query* fixed effect (one dummy per task $\times$ query id), and a finer *query* $\times$ *generator* $\times$ *retriever* fixed effect, since the same query is evaluated under all four generator/retriever cells and a pure query fixed effect can still let cell identity contribute to the “within” variation.

**Interaction and marginal-effects model.** For Finding 4 (Section 5.4), we test whether a single additive PL-vs-MMR coefficient is an adequate summary by adding a  $\text{PL} \times \text{ILD}_{\text{norm}}$  interaction term. When the interaction is non-negligible, we report the conditional PL–MMR gap at several points across the observed  $\text{ILD}_{\text{norm}}$  range, and the average marginal effect (AME) — the interaction model evaluated at the empirical mean of  $\text{ILD}_{\text{norm}}$  — with delta-method standard errors computed from the full coefficient covariance matrix.

### 3.8 Task Weighting

LaMP task sample sizes differ substantially after filtering (Table 2). Pooled query-level regressions and tests throughout this paper weight each *query* equally, not each task; larger tasks therefore contribute proportionally more observations to any pooled estimate. We report task-level heterogeneity and leave-one-task-out results alongside every pooled number specifically so that this implicit weighting does not go unnoticed.

## 4 Experimental Setup

Table 1 summarizes the full factorial design: 2 generators  $\times$  2 retrievers  $\times$  12 reranking settings  $\times$  7 LaMP tasks, for 336 total experiment configurations and 163,392 query-level observations before the normalization exclusion of Section 3.6. Table 2 gives the retained query count per task and generator after the source paper’s per-generator utility-label filtering.

Table 1: Experimental design.

| Component                        | Values                                                               |
|----------------------------------|----------------------------------------------------------------------|
| Benchmark                        | LaMP, all 7 tasks, “user” split                                      |
| Generators                       | Flan-T5-Small (77M), Flan-T5-Base (250M)                             |
| Retrievers                       | BM25, Contriever                                                     |
| Candidate list size              | $k = 5$                                                              |
| Deterministic baseline           | top- $k$ by raw retrieval score                                      |
| MMR                              | $\lambda \in \{0.15, 0.30, 0.45, 0.55, 0.70, 0.85, 1.00\}$           |
| PL                               | $\alpha \in \{1, 2, 4, 8\}$ , $N = 10$ sampled rankings per $\alpha$ |
| Settings per cell                | 12 (1 deterministic + 7 MMR + 4 PL)                                  |
| Total configurations             | 336                                                                  |
| Query-level rows (pre-exclusion) | 163,392                                                              |

**Sanity check: MMR at  $\lambda = 1$ .** By construction,  $\lambda = 1$  removes MMR’s diversity term entirely and should reproduce the deterministic baseline. We verify this directly rather than assuming it: the maximum paired EU difference between the two conditions is **exactly 0.0 across all 13,616 paired queries**. This confirms outcome equality, not identical document rankings, which we did not independently verify at the document-ID level.

### 4.1 Premise Check: Does PL’s Fairness Mechanism Actually Increase Diversity Here?

The motivating confound requires that PL’s fairness intervention actually moves diversity as a byproduct; we test this rather than assume it. Figure 1 shows both quantities as PL’s  $\alpha$  varies,

Table 2: Retained query count per LaMP task and generator, after the source paper’s per-generator utility-label filtering.

| LaMP task | Flan-T5-Base | Flan-T5-Small |
|-----------|--------------|---------------|
| 1         | 232          | 51            |
| 2         | 280          | 192           |
| 3         | 378          | 311           |
| 4         | 827          | 833           |
| 5         | 759          | 826           |
| 6         | 783          | 760           |
| 7         | 211          | 365           |

averaged across the four (generator, retriever) cells.

**Query-level coupling is weak.** Within PL rows, the Pearson correlation between  $EE-D_{\text{norm}}$  and  $ILD_{\text{norm}}$  is  $r = -0.086$  ( $n = 54,464$ ) — a small effect once individual queries are considered, swamped by per-query noise in both quantities.

**The aggregate, by- $\alpha$  pattern is monotonic.** Table 3 reports the mean source-normalized  $EE-D$  and mean raw  $ILD$ -Jaccard at each  $\alpha$ : as  $\alpha$  increases (fairness intervention weakens toward the deterministic baseline), disparity rises and diversity falls, monotonically in the expected directions.

Table 3: Aggregate fairness and diversity by PL  $\alpha$  (lower  $\alpha$  is fairer).

| PL $\alpha$ | Mean source-normalized $EE-D$ | Mean $ILD$ -Jaccard (raw) |
|-------------|-------------------------------|---------------------------|
| 1           | 0.210                         | 0.879                     |
| 2           | 0.248                         | 0.870                     |
| 4           | 0.657                         | 0.838                     |
| 8           | 0.948                         | 0.830                     |

The premise therefore holds *in aggregate, as a tendency across the fairness dial*, but not as a tight per-query coupling. We report both because this looseness matters downstream: it is part of why a PL-vs-MMR comparison adjusted for measured diversity (Section 5.4) cannot be treated as a clean causal isolation of fairness’s own contribution.

## 5 Results

### 5.1 Finding 1: PL Reranking Generally Does Not Improve Utility

We test each PL  $\alpha$  against the deterministic baseline, paired per-query, Holm-corrected within each cell’s 4-test family ( $4 \alpha$  values  $\times$  4 generator/retriever cells = 16 tests total). This scopes the test specifically to PL, since MMR has no fairness objective and does not belong in a test about whether fairness pays off; MMR-vs-baseline comparisons are evidence about diversity instead (Section 5.3).

Of the 16 tests, **10 are statistically significant, and 8 of those are losses**. The two significant gains are both the same PL configuration ( $\alpha = 4$ ) in two different cells, and are small ( $+0.0014$  and  $+0.0025$  in  $EU_{\text{norm}}$ ) against losses roughly an order of magnitude larger elsewhere.

Kim and Diaz’s own reported result is more nuanced than “fair rankings often outperform”: they report an overall fairness/quality tradeoff in which moderate fairness settings sometimes preserve

Fairness interventions inherently raise diversity: PL moves both dials at once, MMR moves only diversity

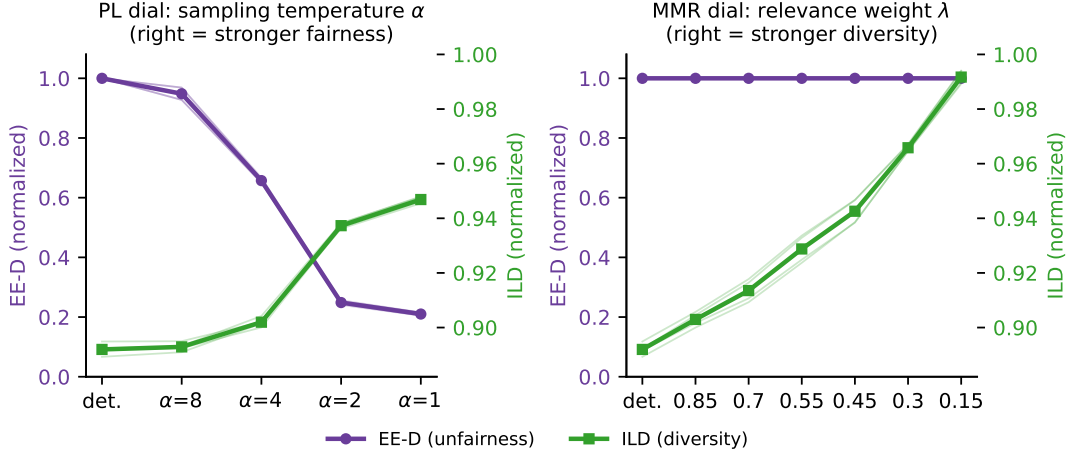


Figure 1: Fairness interventions raise diversity in aggregate: PL’s dial moves both expected exposure disparity and diversity at once (left panel), while MMR’s dial moves diversity with disparity pinned near the deterministic level (right panel, EE-D  $\approx 1$ ), which is what makes MMR usable as a diversity-only instrument. Bold lines are the mean across the four (generator, retriever) cells; thin lines are individual cells.

or improve utility. Given that our evaluation covers 2 of their 4 generators, 2 of their 3 retrievers, and  $N = 10$  sampled rankings against their  $N = 100$ , we do not frame this as contradicting their result outright. Within the two smaller generators and two retrievers evaluated here, however, we do not reproduce the pattern that moderate fair reranking frequently preserves or improves deterministic-baseline utility: the direction is overwhelmingly negative, with two small, real, but practically negligible exceptions.

## 5.2 Finding 2: A Real Fairness–Utility Tradeoff Within PL’s Own Fairness Dial

We regress  $EU_{\text{norm}}$  on  $EE-D_{\text{norm}}$ , restricted to PL rows only, controlling for  $ILD_{\text{norm}}$  and generator/retriever indicators:

$$EU_{\text{norm}}(q, s) = \beta_0 + \beta_1 EE-D_{\text{norm}}(q, s) + \beta_2 ILD_{\text{norm}}(q, s) + \beta_3 \mathbb{I}[\text{Base}] + \beta_4 \mathbb{I}[\text{Contriever}] + \varepsilon(q, s),$$

$s \in \text{PL rows.} \quad (1)$

Restricting to PL rows avoids mixing a within-method fairness manipulation (PL’s  $\alpha$ ) with a between-method contrast (MMR and the deterministic baseline sit at a fixed EE-D  $\approx 1$  regardless of  $\lambda$ ), which would not be a clean fairness-dial estimate.

The primary estimate is  $\hat{\beta}_1 = +0.0735$  (cluster-robust  $p = 6.3 \times 10^{-54}$ ,  $n = 52,568$ , 4,269 query clusters; the naive, non-clustered  $p = 2.1 \times 10^{-93}$  overstates confidence by treating repeated observations of the same query as independent). Because lower EE-D means fairer, a *positive* coefficient means that **less fair, more disparate rankings are associated with higher utility** — the opposite of what a “fairness helps” story would predict.

**Robustness.** Leave-one-task-out, on the identical specification and cluster-robust inference, keeps the coefficient positive and highly significant under every single-task exclusion, ranging from

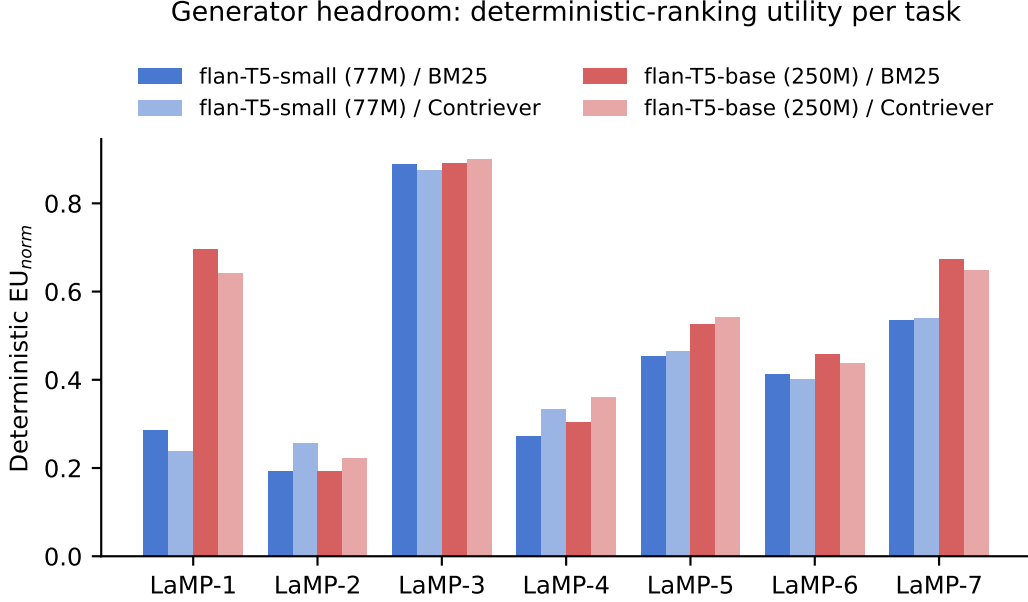


Figure 2: Deterministic-baseline generation quality ( $EU_{\text{norm}}$ ) per LaMP task, for all four (generator, retriever) cells. LaMP tasks differ substantially in achievable utility even after normalization, motivating the per-query normalization used throughout (Section 3.6).

+0.0613 (dropping LaMP-4,  $p = 1.9 \times 10^{-27}$ ) to +0.0834 (dropping LaMP-5,  $p = 2.7 \times 10^{-53}$ ); no sign or significance flips. Query fixed effects — isolating the within-query association between  $EE-D_{\text{norm}}$  and  $EU_{\text{norm}}$  from between-query composition — give +0.0518 ( $p = 2.0 \times 10^{-30}$ ); the finer query $\times$ generator $\times$ retriever fixed effect, which isolates purely the  $\alpha$ -driven variation for one exact query in one exact cell, gives +0.0526 ( $p = 2.4 \times 10^{-30}$ ), essentially unchanged from the coarser fixed effect. The pooled estimate was picking up some between-query variation, but the within-query association remains substantial under every specification we tested (Table 8).

**Task heterogeneity.** Fit on each task alone (no leave-one-task-out exclusion, no other controls), the coefficient is positive in all seven tasks, but not statistically significant on two of them in isolation (LaMP-3,  $p = 0.31$ ; LaMP-7,  $p = 0.71$ ); the remaining five are individually significant and positive. Unlike Findings 3 and 4, no task individually reverses this association’s sign. We nonetheless describe this as a robust *pooled* association rather than a claim that it is statistically detectable within every single task in isolation, since two of the seven do not reach significance on their own.

### 5.3 Finding 3: No General Utility Benefit From MMR Diversity

If diversity were secretly responsible for any apparent fairness benefit, diversity should help on its own, holding fairness constant — exactly what MMR, which has no fairness mechanism, lets us test. We regress  $EU_{\text{norm}}$  on  $ILD_{\text{norm}}$  within MMR rows only (exposure disparity is fixed near the deterministic level,  $EE-D \approx 1$ , for all MMR settings):

$$EU_{\text{norm}}(q, s) = \gamma_0 + \gamma_1 ILD_{\text{norm}}(q, s) + \gamma_2 \mathbb{I}[\text{Base}] + \gamma_3 \mathbb{I}[\text{Contriever}] + \varepsilon(q, s),$$

$s \in \text{MMR rows.} \quad (2)$

Overview: fairness vs. utility, all methods (per-setting means) - PL's  $\alpha$ -trajectory highlighted; the statistical test (Finding 2) uses PL rows only

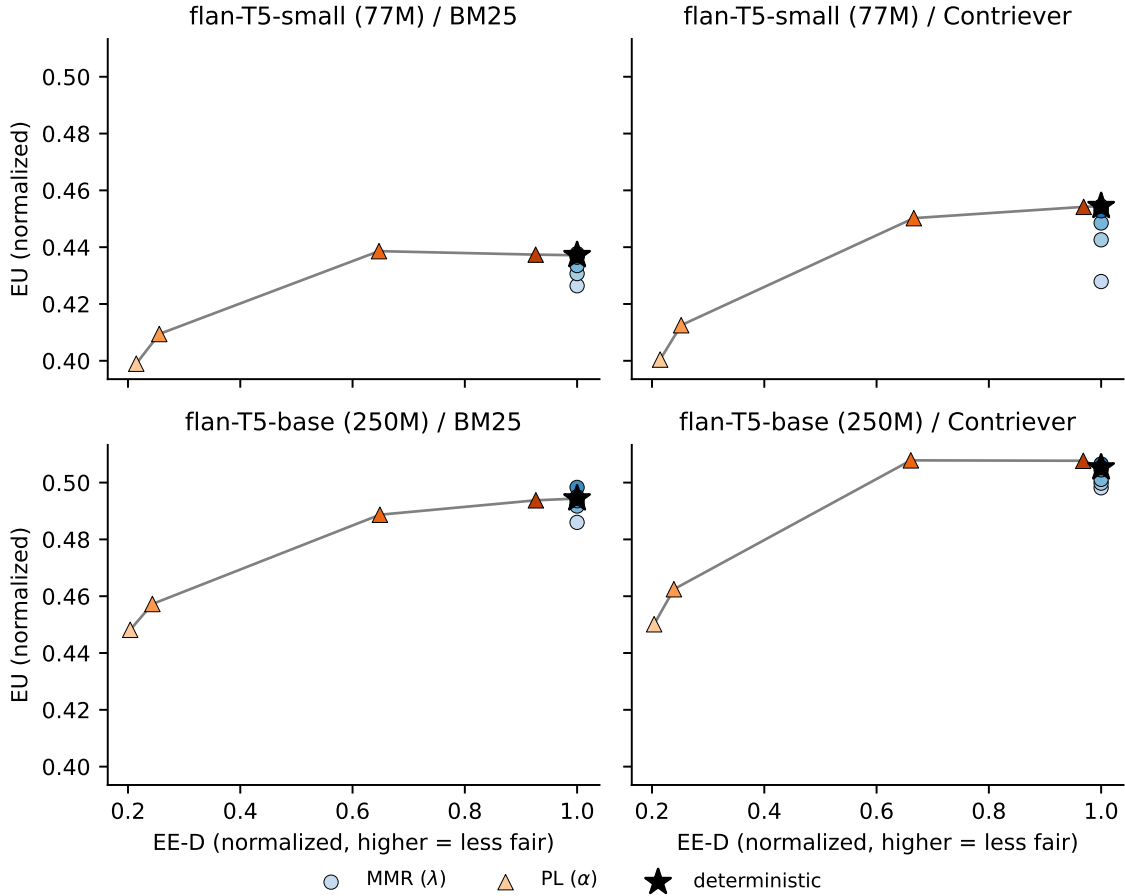


Figure 3: Fairness ( $EE-D_{\text{norm}}$ , higher is less fair) versus utility ( $EU_{\text{norm}}$ ), per-setting means, one panel per generator/retriever cell. PL's  $\alpha$ -trajectory is highlighted (triangles); MMR (circles) and the deterministic baseline (star) are shown for context, but the Finding 2 regression (Eq. 1) uses PL rows only.

We find **no evidence of a general utility benefit from increased MMR diversity**: the pooled association is modestly negative,  $\hat{\gamma}_1 = -0.0675$  ( $n = 91,994$ ), cluster-robust  $p = 0.0115$  — real, but a considerably more modest significance level than the naive  $p = 5.6 \times 10^{-10}$  suggested, since the repeated-query structure was inflating confidence. We treat  $p = 0.0115$  as modest evidence for a real but small effect, not overwhelming evidence.

**Robustness.** The pooled coefficient stays negative under every single-task exclusion (a sign check; no leave-one-task-out significance claim is at stake here), but it does not hold in every task individually: fit on LaMP-2 or LaMP-5 alone, the coefficient is positive. The pooled negative association is robust to removing any one task, while the direction itself is heterogeneous across tasks — diversity’s effect looks benchmark-dependent, not a fixed property of RAG generation. Query fixed effects give  $-0.0820$  ( $p = 2.4 \times 10^{-4}$ ), if anything larger in magnitude than the pooled estimate; the finer query  $\times$  generator  $\times$  retriever fixed effect gives  $-0.0785$  ( $p = 5.0 \times 10^{-4}$ ), essentially unchanged (Table 8).

**Per-cell heterogeneity.** Figure 4 and Table 4 report the identical bivariate specification (ILD<sub>norm</sub> only, no controls) fit separately within each cell, with cluster-robust inference. Only the Flan-T5-Base/BM25 cell is individually significant. A formal test of whether the visually larger Base-cell coefficients reflect a genuine generator-size interaction — a pooled within-MMR regression with an ILD<sub>norm</sub>  $\times$  generator interaction term, cluster-robust by query and controlling for task and retriever — gives  $p = 0.28$ : not significant. The descriptive per-cell pattern in Figure 4 is real and visible, but the formal test does not confirm generator size as its driver.

Table 4: Per-cell bivariate regression of EU<sub>norm</sub> on ILD<sub>norm</sub> within MMR rows (no controls), matching Figure 4 exactly.

| Cell                       | Coefficient | $p$                  |
|----------------------------|-------------|----------------------|
| Flan-T5-Small / BM25       | $-0.0374$   | 0.34                 |
| Flan-T5-Small / Contriever | $-0.0149$   | 0.69                 |
| Flan-T5-Base / BM25        | $-0.1483$   | $8.9 \times 10^{-5}$ |
| Flan-T5-Base / Contriever  | $-0.0728$   | 0.068                |

**Direct MMR-vs-deterministic comparison.** As a complementary, non-regression check, we test whether MMR ever significantly beats or loses to the deterministic baseline directly: paired per-query, all 7  $\lambda$  values  $\times$  4 cells = 28 comparisons (every MMR setting run, not a pre-selected subset), Holm-corrected within each cell’s 7-test family. **4 of 28 are significant, and all 4 are losses** (Table 5). The pattern is consistent with the negative diversity–utility association above (lower  $\lambda$  trades more relevance for more diversity, not diversity in isolation, so this is not independent confirmation of a diversity-specific effect): every significant loss occurs at the two strongest-diversity settings, and the per-cell delta shrinks toward zero monotonically as  $\lambda \rightarrow 1$ , where it is exactly 0.0 in every cell by the construction check of Section 4. Overall, 19 of 28 comparisons point negative, and MMR shows no significant utility *win* over the deterministic baseline in any of the 28 settings tested.

#### 5.4 Finding 4: A Residual PL–MMR Utility Difference After Adjusting for Diversity

Restricting to PL and MMR rows, we ask whether being a PL setting (versus MMR) predicts a further utility difference once measured diversity is accounted for. This is a regression adjustment, not a matching procedure.

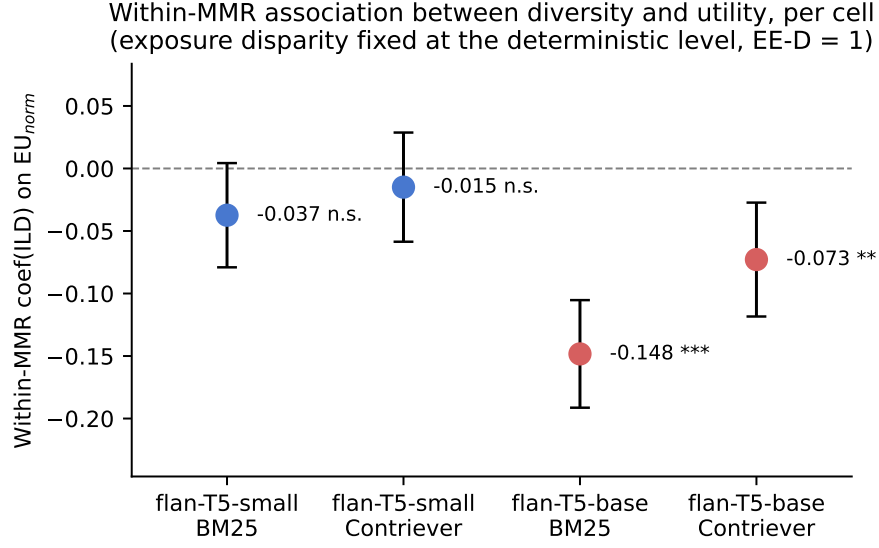


Figure 4: Within-MMR association between diversity and utility, per cell, with exposure disparity fixed at the deterministic level. Coefficients correspond to Table 4.

Table 5: MMR versus deterministic baseline, paired per-query, aggregated over the four generator/retriever cells (28 comparisons total, Holm-corrected within each cell’s 7-test family).

| MMR $\lambda$ | Cells with negative $\Delta$ (of 4) | Cells significant (of 4) |
|---------------|-------------------------------------|--------------------------|
| 0.15          | 4                                   | 3                        |
| 0.30          | 4                                   | 1                        |
| 0.45          | 4                                   | 0                        |
| 0.55          | 2                                   | 0                        |
| 0.70          | 3                                   | 0                        |
| 0.85          | 2                                   | 0                        |
| 1.00          | 0 (exactly zero, by construction)   |                          |

**Additive specification.** Table 6 reports the per-cell coefficient of a PL indicator on  $EU_{\text{norm}}$ , controlling for  $ILD_{\text{norm}}$ ; all four cells are negative, significant, and of similar magnitude. Pooling with a single common  $ILD_{\text{norm}}$  slope for PL and MMR gives  $\hat{\beta}_1 = -0.0194$  (cluster-robust  $p = 8.8 \times 10^{-34}$ ) — but before treating this single number as the answer, we test whether a single common slope is the right model.

Table 6: Per-cell coefficient of the PL indicator on  $EU_{\text{norm}}$ , controlling for  $ILD_{\text{norm}}$  (additive specification).

| Cell                       | Coefficient | $p$                   |
|----------------------------|-------------|-----------------------|
| Flan-T5-Small / BM25       | −0.0130     | $1.1 \times 10^{-4}$  |
| Flan-T5-Base / BM25        | −0.0226     | $8.2 \times 10^{-11}$ |
| Flan-T5-Small / Contriever | −0.0186     | $4.7 \times 10^{-8}$  |
| Flan-T5-Base / Contriever  | −0.0227     | $1.3 \times 10^{-10}$ |

**Primary specification: does the gap depend on diversity level?** We add a  $PL \times ILD_{\text{norm}}$  interaction term:

$$EU_{\text{norm}}(q, s) = \beta_0 + \beta_1 \mathbb{K}[\text{PL}](s) + \beta_2 ILD_{\text{norm}}(q, s) + \beta_3 (\mathbb{K}[\text{PL}](s) \times ILD_{\text{norm}}(q, s)) + \text{controls} + \varepsilon(q, s). \quad (3)$$

The interaction is significant in the pooled specification ( $\hat{\beta}_3 = +0.0293$ ,  $p = 0.0125$ ): the two methods’  $ILD_{\text{norm}}$  slopes are not identical, so a single additive coefficient understates what is happening. Because  $ILD_{\text{norm}}$  is not centered,  $\hat{\beta}_1$  alone is the model’s fitted PL–MMR gap specifically at  $ILD_{\text{norm}} = 0$ ; that value does occur in the data (both methods’ observed  $ILD_{\text{norm}}$  minimum is exactly 0.0) but is rare, since  $ILD_{\text{norm}}$ ’s own 10th percentile is already 0.84. Table 7 reports the conditional gap,  $\Delta(ILD_{\text{norm}}) = \hat{\beta}_1 + \hat{\beta}_3 \cdot ILD_{\text{norm}}$ , at several points across the observed range, together with the average marginal effect (AME), the same quantity evaluated at the empirical mean of  $ILD_{\text{norm}}$ .

Table 7: Conditional PL–MMR gap at points across the observed  $ILD_{\text{norm}}$  range, and the average marginal effect (AME), from the interaction model (Eq. 3).

| $ILD_{\text{norm}}$ level | $\Delta(\text{PL} - \text{MMR})$ | 95% CI           | $p$                   |
|---------------------------|----------------------------------|------------------|-----------------------|
| 0 (observed floor, rare)  | −0.0464                          | [−0.068, −0.025] | $2.1 \times 10^{-5}$  |
| $p_{10}$ (0.84)           | −0.0218                          | [−0.025, −0.018] | $2.5 \times 10^{-32}$ |
| median (0.96)             | −0.0181                          | [−0.021, −0.015] | $2.8 \times 10^{-27}$ |
| $p_{90}$ (0.99)           | −0.0172                          | [−0.021, −0.014] | $3.6 \times 10^{-21}$ |
| <b>mean (AME)</b>         | −0.0192                          | [−0.022, −0.016] | $1.5 \times 10^{-33}$ |

The gap is largest, about −0.046, in the small, rare tail of low-diversity settings, and narrows to roughly −0.017 to −0.022 across the bulk of the data ( $p_{10}$  to  $p_{90}$  of observed  $ILD_{\text{norm}}$ ). We treat the AME, −0.0192, as the headline single-number summary of Finding 4: a proper average over the interaction model rather than an assumption of one fixed slope, and it lands almost exactly where the simpler additive coefficient (−0.0194) did.

**Is the interaction itself robust?** Not entirely, though the AME built from it is. Under the strictest identification we use — query×generator×retriever fixed effects — the interaction shrinks

and loses significance ( $\hat{\beta}_3 = +0.0157$ ,  $p = 0.15$ ), about half the pooled magnitude and no longer distinguishable from zero. Leave-one-task-out on the pooled interaction model keeps the coefficient positive when 5 of 7 tasks are dropped individually (range  $+0.023$  to  $+0.037$ ,  $p$  from 0.003 to 0.06), but **flips sign when LaMP-6 is dropped** ( $-0.0280$ ,  $p = 0.16$ , itself not significant either way). We therefore treat the diversity-dependence pattern — the interaction term itself — as a suggestive, secondary observation, not an established one. What *does* survive, robustly, across every specification we tried (pooled, query fixed effects, query $\times$ generator $\times$ retriever fixed effects, and all seven leave-one-task-out folds) is the AME itself: it ranges narrowly from  $-0.0161$  to  $-0.0217$  across every leave-one-task-out fold (all  $p < 10^{-17}$ ), and from  $-0.0192$  to  $-0.0200$  across the pooled and fixed-effects variants (Table 8), always highly significant and never close to flipping sign. Figure 5 shows the raw per-quintile pattern that motivates the interaction check; we caution against reading the apparent low-vs-high-diversity gap in that figure as an established, robust pattern in its own right — it is the AME in Table 8, not the bin-level shape of Figure 5, that is the robust quantity.

**What this does and does not establish.** Adjusting for  $ILD_{\text{norm}}$  controls for one measured outcome of the ranking, not for everything that differs between PL and MMR: PL is stochastic and MMR deterministic, the two methods may preserve the underlying relevance ordering differently, and PL introduces variance across repeated draws that MMR does not have. The AME is therefore an *average adjusted difference* between PL and MMR after controlling for measured  $ILD_{\text{norm}}$ , not a causal estimate of what fairness itself costs.

**Common support.** Does the comparison rely on extrapolating outside the diversity range either method actually reaches? No: PL’s and MMR’s measured  $ILD_{\text{norm}}$  distributions essentially fully overlap (matched 5th/median/95th percentiles), and restricting to their common support changes nothing ( $\hat{\beta}_1 = -0.0193$ ,  $p = 8.8 \times 10^{-34}$ ; all 144,562 rows already fall inside the overlap).

**Mechanism test.** We hypothesized that PL pays a *relevance* cost — stochastic sampling moves away from the highest-ranked item, and that specifically, not diversity, explains the residual. EE-R is a direct proxy for exposure-relevance alignment. Adding EE-R $_{\text{norm}}$  as a covariate moves  $\hat{\beta}_1$  from  $-0.0194$  to  $-0.0187$  ( $p = 3.0 \times 10^{-34}$ ) — essentially unchanged, even though EE-R $_{\text{norm}}$  itself is strongly associated with utility on its own ( $\hat{\beta} = +0.158$ ,  $p = 5.1 \times 10^{-104}$ , a sanity check that EE-R $_{\text{norm}}$  behaves as expected). **This relevance-cost hypothesis is not confirmed by the test.** PL underperforms MMR on average after adjusting for  $ILD_{\text{norm}}$ , robustly across the specifications tested; the pooled interaction model suggests a larger disadvantage in the rare low-diversity tail, but that interaction does not survive the strictest fixed-effects and leave-one-task-out checks, and the mechanism for the residual remains open. Plausible candidates include the stochasticity itself (variance, not just a mean shift) or something  $ILD_{\text{norm}}$  and EE-R $_{\text{norm}}$  together do not capture, but we have not demonstrated which.

## 6 Robustness Analyses

**Leave-one-task-out.** The headline pooled coefficients for Findings 2 and 3 were checked leave-one-task-out; neither’s main conclusion depends on any single task remaining in the sample. Finding 4’s AME is similarly leave-one-task-out robust (every fold  $-0.016$  to  $-0.022$ , always significant), while the PL $\times$  $ILD_{\text{norm}}$  interaction underlying its secondary diversity-dependence framing is not (it flips sign when LaMP-6 is dropped). This does not cover every secondary specification in this paper (per-cell regressions, the generator interaction, the EE-R mechanism check); those were not individually re-run leave-one-task-out.

**Cluster-robust inference.** Added throughout to correct for repeated observations of the same query across settings; it changes exact significance levels (most notably Finding 3, where the

Utility by diversity level: MMR vs PL vs the original deterministic list  
(task-matched bins: only (task, bin) cells where both methods have data)

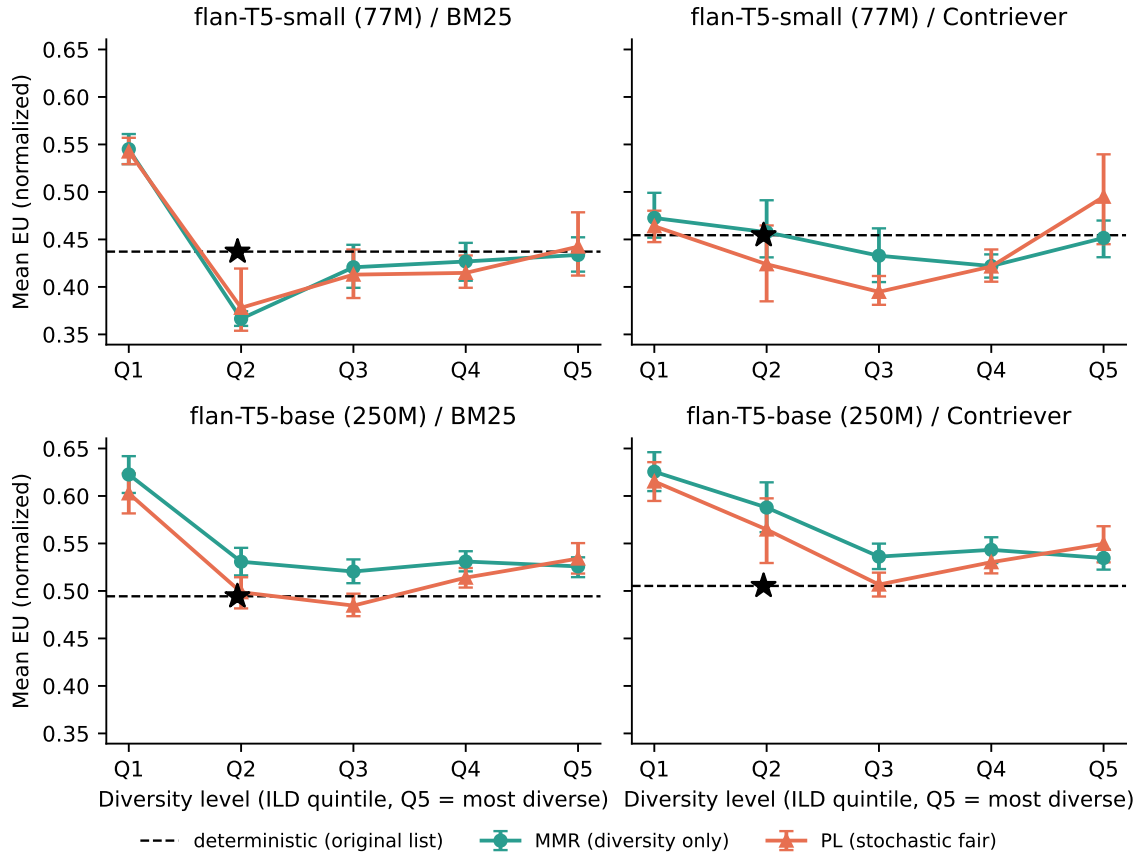


Figure 5: Mean utility by diversity quintile, MMR versus PL, with the deterministic baseline shown for reference. This raw, descriptive pattern motivated the interaction check of Table 7; as discussed in the text, the AME is the robust summary of Finding 4, not the specific low-vs-high quintile shape visible here.

naive  $p = 5.6 \times 10^{-10}$  becomes  $p = 0.0115$  once clustering is applied) but does not overturn any finding’s direction or overall conclusion.

**Fixed effects.** Table 8 summarizes all three findings under query fixed effects and the finer query×generator×retriever fixed effect. For Finding 4, we report the AME from the interaction model at each fixed-effects level (not the additive coefficient) as the headline robustness quantity, per the primary specification of Section 5.4; the additive coefficient (also fixed-effects-robust, at  $-0.0200$  under both levels) is a secondary check. The two fixed-effects levels barely differ from each other for any finding, indicating that cell composition was not doing hidden work in the coarser, query-only version.

Table 8: Pooled coefficients versus query fixed effects and the finer query×generator×retriever fixed effect. Finding 4’s row reports the AME from the interaction model (Eq. 3), not the additive coefficient, consistent with Section 5.4.

| Finding                 | Pooled                                   | Query FE                                 | Query×gen.×retr. FE                      |
|-------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| 2 (coef. EE-D, PL rows) | +0.0735<br>( $p = 6.3 \times 10^{-54}$ ) | +0.0518<br>( $p = 2.0 \times 10^{-30}$ ) | +0.0526<br>( $p = 2.4 \times 10^{-30}$ ) |
| 3 (coef. ILD, MMR rows) | −0.0675<br>( $p = 0.0115$ )              | −0.0820<br>( $p = 2.4 \times 10^{-4}$ )  | −0.0785<br>( $p = 5.0 \times 10^{-4}$ )  |
| 4 (AME, PL+MMR)         | −0.0192<br>( $p = 1.5 \times 10^{-33}$ ) | −0.0200<br>( $p = 1.6 \times 10^{-36}$ ) | −0.0200<br>( $p = 2.0 \times 10^{-36}$ ) |

**Normalization sensitivity.** As described in Section 3.6, filling the 3.5% of rows excluded by the zero-denominator condition, rather than dropping them, leaves all three conclusions unchanged.

**Task-robustness figure.** Figure 6 visualizes the leave-one-task-out check for all three pooled additive coefficients together; all three retain their sign and significance status under every single-task exclusion.

## 7 Discussion

This paper set out to answer a specific question: when fair reranking appears to help or hurt RAG generation quality, is that attributable to fairness itself, or to the within-list diversity the fairness mechanism happens to induce? Across the settings we test, the answer is that **measured within-list diversity (ILD-Jaccard) alone does not account for the difference between PL and MMR utility**. PL’s own fairness dial is associated with a real, robust, and negative relationship to utility (Finding 2); MMR’s diversity dial shows no general utility benefit of its own, and if anything a modest negative association (Finding 3); and PL retains an average utility disadvantage relative to MMR even after adjusting for measured diversity (Finding 4). If diversity were the hidden mechanism behind an apparent fairness benefit, MMR’s explicit diversity manipulation should have shown a positive utility association on its own; it does not.

This is a narrower conclusion than Kim and Diaz’s original finding might suggest, but the two results are not necessarily in tension: our evaluation covers 2 of their 4 generators, 2 of their 3 retrievers, and a tenfold-lower PL sample count ( $N = 10$  versus  $N = 100$ ), and we do not claim to have reproduced their full experimental grid. What we can say is that, within the Flan-T5-Small and Flan-T5-Base, BM25 and Contriever setting we do test, fair reranking does not reproduce the pattern of frequently preserving or improving utility, and that the diversity-as-mechanism hypothesis specifically does not explain the residual gap between the two reranking families.

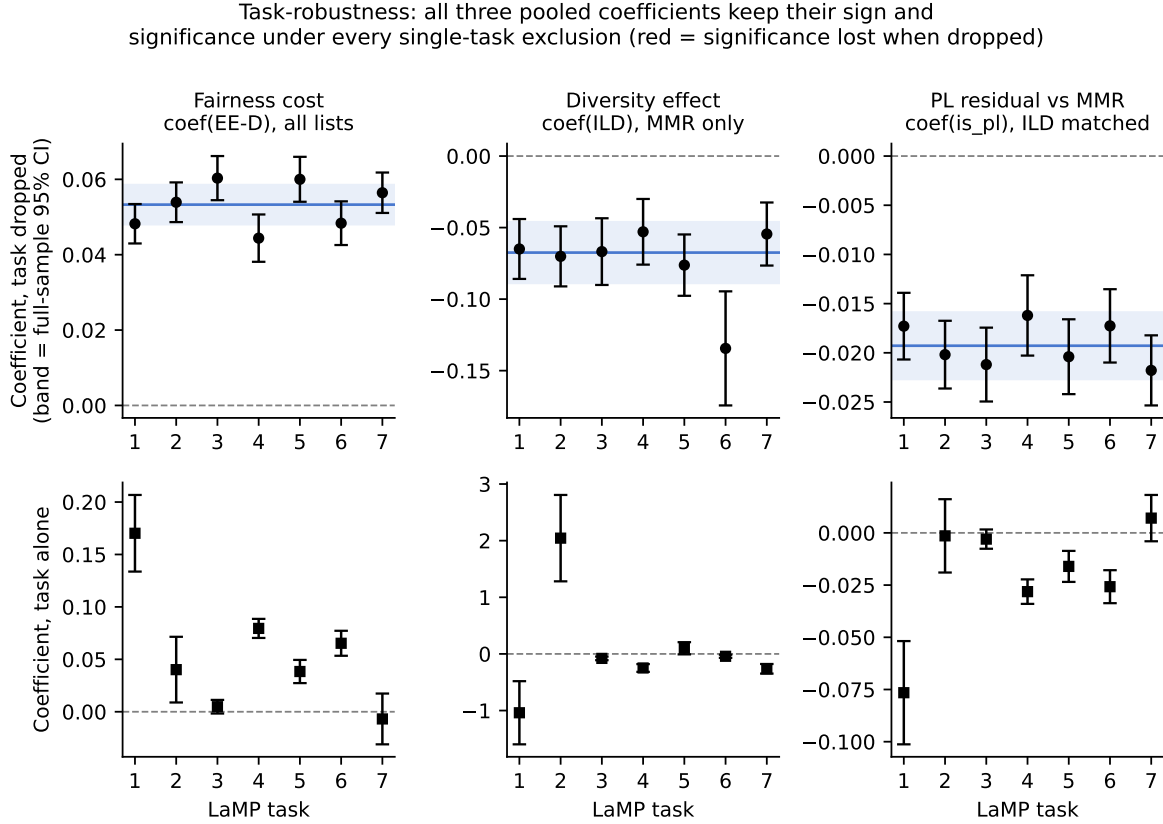


Figure 6: Leave-one-task-out robustness for the three pooled additive coefficients (Findings 2–4). Top row: coefficient when each task is dropped, against the full-sample estimate (blue band, 95% CI). Bottom row: coefficient fit on each task alone. No exclusion flips sign or significance status for any of the three pooled coefficients.

It is important to be precise about what the PL-vs-MMR comparison in Finding 4 does and does not identify. Because PL and MMR differ along several dimensions besides measured diversity — stochastic sampling versus deterministic selection, and how each method preserves or disturbs the underlying relevance ordering, among others — the residual utility gap after adjusting for  $ILD_{\text{norm}}$  is an *adjusted association*, not a causal estimate of what fairness itself costs. We tested the most obvious candidate explanation directly: that PL’s randomization imposes a relevance cost, measurable via exposure-relevance alignment (EE-R). Adding EE-R as a covariate left the residual essentially unchanged, so this specific mechanism is not confirmed. The true mechanism — most plausibly related to the variance PL’s stochastic sampling introduces, rather than only its mean shift, or to some property of the ranking that neither  $ILD_{\text{norm}}$  nor  $EE-R_{\text{norm}}$  captures — remains an open question for future work, which we discuss should require a design that varies fairness and diversity independently rather than only adjusting for one while measuring the other.

Finally, task heterogeneity recurs throughout: Finding 2’s association reverses sign on LaMP-7 alone; Finding 3’s reverses on LaMP-2 and LaMP-5 alone; Finding 4’s secondary diversity-dependence interaction reverses sign when LaMP-6 is excluded. None of these task-level reversals overturn the corresponding pooled, leave-one-task-out-robust conclusion, but they are a reminder that these associations are properties of a query-weighted pool of seven quite different personalization tasks, not of RAG generation in general.

## 8 Limitations

- **Generator scale.** We evaluate only Flan-T5-Small (77M) and Flan-T5-Base (250M). Whether this tradeoff grows, shrinks, or reverses at larger generator scales is untested here; utility labels exist for a much larger (XXL) generator in the source release, but it does not fit the GPU hardware available for this project.
- **Retriever coverage.** We evaluate BM25 and Contriever only, 2 of the 3 retrievers in the source paper’s evaluation.
- **PL sample count.**  $N = 10$  sampled rankings per  $(q, \alpha)$  pair, against the source paper’s  $N = 100$ ; our PL-based EE-D/EU estimates carry more Monte Carlo noise than theirs. A cheap follow-up — recomputing EE-D only at  $N = 100$  on a subset, without regenerating text — would bound how much this matters, but was not run for this paper.
- **Diversity operationalization.** We measure diversity as ILD-Jaccard, a single specific notion of within-list textual variety; our conclusion that measured diversity does not account for the PL–MMR gap is a claim about *this* operationalization, not about every possible notion of diversity.
- **Loose fairness–diversity coupling.** EE-D and ILD are conceptually distinct quantities that are only loosely, not strictly, coupled under PL in this data (Section 4.1); this limits how cleanly any  $ILD_{\text{norm}}$ -adjusted comparison can isolate fairness’s own contribution.
- **PL and MMR differ beyond diversity.** PL is stochastic and MMR deterministic; the two methods may preserve the original relevance ordering differently, and only PL introduces sampling variance across repeated draws. Finding 4’s residual should not be read as an isolated measurement of fairness’s causal effect for this reason.
- **Task dependence.** Findings 2 and 3’s pooled effects are leave-one-task-out robust, but individual tasks (LaMP-7 for the fairness–utility tradeoff; LaMP-2 and LaMP-5 for the diversity

association) go the opposite direction on their own. Finding 4’s AME is similarly robust, but the interaction term underlying its secondary diversity-dependence framing is not (it loses significance under the strictest fixed effects and flips sign when LaMP-6 is dropped); we treat that pattern as secondary for exactly this reason.

## 9 Conclusion

We asked whether an apparent fairness benefit in RAG generation quality is attributable to fairness itself or to the within-list diversity that a fairness mechanism happens to induce. On a full-coverage evaluation across all seven LaMP tasks, two generators, and two retrievers, we find that PL fair reranking does not generally improve utility over a deterministic baseline; that within PL’s own fairness dial, greater exposure equality is robustly associated with lower utility; that MMR’s explicit diversity dial shows no general utility benefit on its own; and that PL retains an average utility disadvantage relative to MMR even after adjusting for measured diversity, a disadvantage that a direct test of a relevance-cost mechanism does not explain. Taken together, **measured within-list diversity (ILD-Jaccard) alone does not account for the PL–MMR utility difference** we observe, while the residual itself should not be read as the isolated causal cost of fairness, since PL and MMR differ along dimensions besides diversity. Determining what does explain that residual — and whether these patterns persist at larger generator scale and across additional retrievers — is the natural next step, and would benefit from an experimental design that varies fairness and diversity independently rather than adjusting for one while measuring the other.

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## A Supplementary Figure

PL vs MMR utility at matched diversity, per task: small, mostly negative residuals

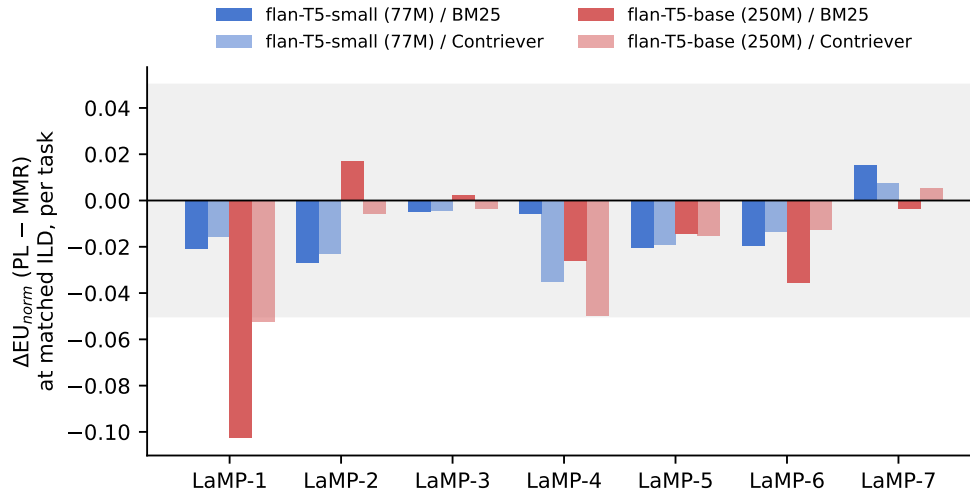


Figure 7: Per-task detail underlying Finding 4: mean  $EU_{\text{norm}}$  difference (PL minus MMR) at diversity-matched bins, by LaMP task and generator/retriever cell. Residuals are small and mostly negative, consistent with the pooled AME of Section 5.4; this per-task breakdown is supplementary detail, not an independent robustness claim.